

Venus

With the upcoming shuttle flight, Atlantis, which will deploy the Magellan Probe to Venus on April 28th, this might be a good time to review our current knowledge of Venus.

Venus is the second planet in distance from the Sun. This neighbor of Earth is very similar in such gross characteristics as mass, radius, and density. In other ways, Venus is apparently different. Its atmospheric mass is almost one hundred times that of Earth. Its atmosphere is mostly carbon dioxide (about 96%), the balance being mostly nitrogen. An extensive cloud layer of concentrated sulphuric acid is present. Its surface temperature is an unbearable 750 degrees Kelvin. Venus rotates with a period of 243 days, from east to west, in the opposite direction of most other planets.

To the naked eye, Venus is the brightest starlike object in the sky. Like the Moon, Venus ranges over a full set of phases, from a "new moon" to a "full moon".

The light seen coming from Venus is almost entirely due to sunlight that is reflected from a dense cloud layer. Its water content is much less than that of Earth.

The high value of Venus's surface temperature is not due to its being close to the sun. Because its cloud layer reflects back to space about 75% of the incident sunlight, Venus absorbs less solar energy than does the Earth. Rather, the high temperature is the result of a very efficient greenhouse effect. Venus allows a small but significant fraction of the incident sunlight to penetrate the surface but prevents a large fraction of the heat generated from escaping directly to space.

As Venus's axis of rotation is almost perpendicular to its orbital plane, its climate has little seasonal variability.

Venus's interior is quantitatively similar to Earth in having a central iron core, a middle mantle made of rocks rich in silicon. Its core may now be solid which accounts for the absence of a detectable magnetic field.

Observation by radar signals show giant craters, large volcanic constructs, enormous rift valleys and jagged mountain ranges. The radar observations were done by the Venera spacecraft.

The current high surface temperature and acid constitution precludes the existence of living organisms like those that inhabit the Earth.

Observations of Venus have been carried out by the United States Mariner spacecraft, by the Pioneer Venus spacecraft which consisted of an orbiter and five probes that descended into its atmosphere, and by a number of Soviet Venera spacecraft.

(by Harold Moorin)

A calendar of astronomical events for February, taken from the Astronomical Calendar 1989 by Guy Ottewill.

- 1 (3 UT) Mercury 3.9' north of Venus. They are 15' from the sun in the morning sky; magnitudes 1.7 & -3.9. They are closest (3.8') at 17 UT.
- 3 (6 UT) Moon passes 4' south of Uranus.
(15 UT) Moon passes 5' south of Saturn.
(18 UT) Moon passes 5' south of Neptune.
- 4 (3 UT) Comet d'Arrest at perihelion.
(18 UT) Moon passes 6' south of Mercury.
- 5 (14 UT) Mercury stationary in right ascension; resumes direct eastward motion.
- 6 Alpha Aurigid meteors.
(7:36 UT) New Moon.
- 7 (22 UT) Moon at perigee. Distance 56.6 earth-radii.
- 11 (8 UT) The equation of time is at minimum for the year, -14m 18s. That is, when sundial (apparent solar) time is noon, mean solar time is about 14 minutes later.
- 12 (7 UT) Moon passes 4' north of Mars.
(23:14 UT) Moon at first quarter.
- 13 (7 UT) Moon passes 6' north of Jupiter.
- 16 (4 UT) Sun enters Aquarius, at longitude 327.47' on ecliptic.
(23 UT) Spring equinox on Mars.
- 18 (16 UT) Sun enters the astrological sign Pisces; i.e. its longitude is 330'.
- 20 (11 UT) Moon passes 0.02' north of Regulus. Occultation, visible from Japan, north Pacific, Hawaii, southeast Pacific.
(12 UT) Pluto stationary in right ascension; begins retrograde (westward) motion.
(15:30 UT) Full Moon. Called Snow, Hunger or Wolf Moon. Total eclipse of Moon visible in Asia and Australasia.
- 21 (2 UT) Asteroid Juno at Opposition.
- 23 (14 UT) Moon at apogee. Distance 63.6 Earth-radii.
- 24 (2 UT) Venus at aphelion, 0.7283 a.u. from the sun.
- 26 Delta Leonid meteors.
- 28 (7 UT) Comet Perrine-Mrkos possibly at perihelion.
(19 UT) Moon passes 0.7' south of Antares. Occultation visible from south Indian Ocean, Antarctica.
(20:08 UT) Moon at last quarter.

The end of February, on clear moonless nights, is the best time to look for the zodiacal light, from 1 to 2 hours after sunset, around the the part of the ecliptic near the sun; and the Gegenschein or counter glow, around midnight and around the part of the ecliptic high in the sky. They are caused by sunlight reflected from meteoric dust lying in the plane of the solar system and extending far out beyond the earth.

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Lunar Laughs!

Editor's Notes

A recent study conducted by astronomers from the University of Arizona sheds light on why days are shorter in the winter and longer in the summer. Simply, the temperature changes cause expansion and contraction of daylight!

"What can be moved from place to place, made larger or smaller, filled or made empty but can never be touched?"

(a hole " ... a 'hole' lot of thought went into that one."

I would like to start off this year with saying that the PAS-Times is the club's newsletter. It can easily be molded, formed and created by the members of this club. Although geared towards announcing the meetings, the PASTimes does post some interesting articles and information. I would like to see more articles by members of the club. If you have an article that you have condensed, or just cut out, please send it in and I will be glad to print it up. Thanks.

In just two years, the most spectacular event in the recent history of astronomy will occur. A complete sol- "Wait a minute! Two years? Why in the world do we have to worry about it now? Two years is a long time"

Two years is a long time, but it seems every one is getting ready for this event. Hotel plans, plane tickets, reservations and accomodations all go hand-in-hand when preparing for a solar eclipse.

But where to go? If you stay here in the valley, the eclipse will be a minor spectacular with 67.5% obscuration on the 11th of July, 1991. If one were to travel a bit, perhaps make a vacation out of it, to an area near Mazatlan, Mexico, the view of the elipse would be heavenly. The solar eclipse will last for 7 minutes and the path of the shadow will be 256 kilometers wide. If the pocket book is a little more allowing, perhaps Hawaii sounds good with excellent viewing as well.

This solar eclipse will be the longest, until the year 2132, so unless you are a health food fanatic or exercise a lot, you most likely won't be around for another good one.